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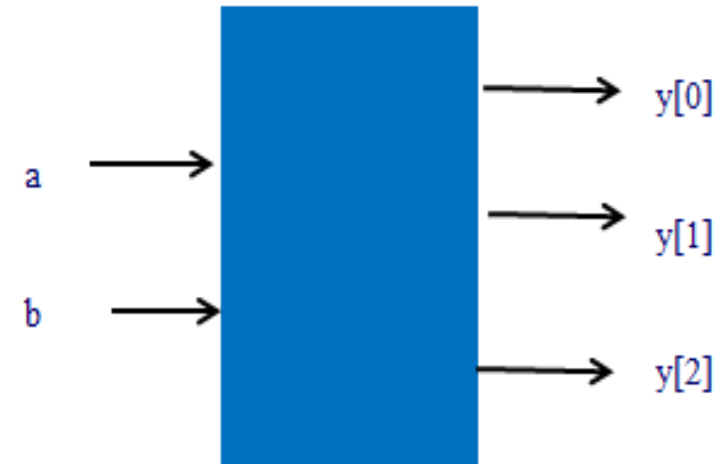
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Example:

Write behavior model of One Bit Comparator

```
module comparator_1_bit(a,b,y);  
input a,b;  
output [0:2]y;  
reg [0:2]y;  
//y[0] = (a=b), y[1] = (a<b) and y[2] = (a>b)  
always @(a,b)  
if (a==b)  
y=3'b100;  
else if (a<b)  
y=3'b010;  
else if (a>b)  
y=3'b001;  
endmodule
```

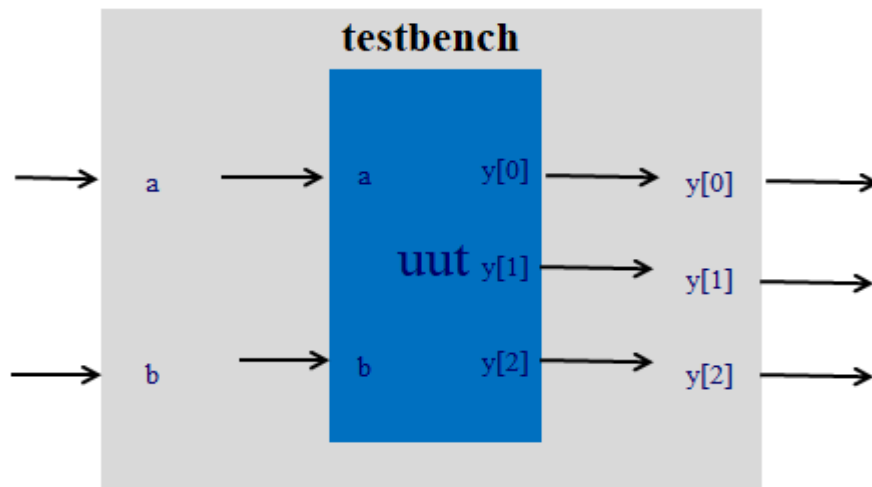
1-Bit Comparator



a	b	y[0]	y[1]	y[2]
0	0	1	0	0
0	1	0	1	0
1	0	0	0	1
1	1	1	0	0

```
module comparator_1_bit(a,b,y);
input a,b;
output [0:2]y;
reg [0:2]y;
//y[0] = (a=b), y[1] = (a<b) and y[2] = (a>b)
always @(a,b)
if (a==b)
y=3'b100;
else if (a<b)
y=3'b010;
else if (a>b)
y=3'b001;
endmodule
```

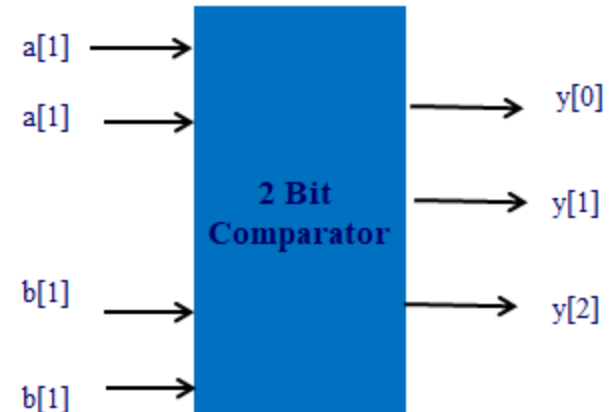
```
module testbench;
reg a; reg b; wire [0:2]y;
comparator_1_bit uut (.a(a), .b(b),.y(y));
initial
begin
$monitor($time,"a=%b,b=%b,y[0]=%b,y[1]=%b,y[2]=%b", a, b,y[0],y[1],y[2]);
#5 a=0; b=0;
#5 a=0; b=1;
#5 a=1; b=0;
#5 a=1; b=1;
#5 $finish;
end
endmodule
```



Example:

Write behavior model of Two Bit Comparator

```
module comparator_2_bit(a,b,y);
input [1:0]a;
input [1:0]b;
output [0:2]y;
reg [0:2]y;
//y[0] = (a=b), y[1] = (a<b) and
y[2] = (a>b)
always @(a,b)
if (a==b)
y=3'b100;
else if (a<b)
y=3'b010;
else if (a>b)
y=3'b001;
endmodule
```



a		b		y[0]	y[1]	y[2]
a[1]	a[0]	b[1]	b[0]			
0	0	0	0	1	0	0
0	0	0	1	0	1	0
0	0	1	0	0	1	0
-	-	-	-	-	-	-
-	-	-	-	-	-	-
-	-	-	-	-	-	-
1	1	1	1	1	0	0

```
module comparator_2_bit(a,b,y);
input [1:0]a;
input [1:0]b;
output [0:2]y;
reg [0:2]y;
//y[0] = (a=b), y[1] = (a<b) and y[2] = (a>b)
always @(a,b)
if (a==b)
y=3'b100;
else if (a<b)
y=3'b010;
else if (a>b)
y=3'b001;
endmodule
```

```
module testbench;
reg [1:0]a; reg [1:0]b; wire [0:2]y;
comparator_1_bit uut (.a(a), .b(b),.y(y));
initial
begin
$monitor($time,"a[1]=%b, a[0]=%b,b[1]=%b,
b[0]=%b,y[0]=%b,y[1]=%b,y[2]=%b", a[1],a[0],
b[1],b[0],y[0],y[1],y[2]);
#5 a=2'b00; b=2'b00;
#5 a=2'b01; b=2'b00;
#5 a=2'b10; b=2'b11;
#5 a=2'b11; b=2'b11;
#5 $finish;
end
endmodule
```

